

Transcriptomic Dissection of Surface Association and Adhesion in *Candida auris*

A Reanalysis of Santana et al. (2023), Science (PRJNA904261)

Total Analysis Cost: \$17.31

1. Executive Summary

This report details the differential gene expression reanalysis of raw paired-end RNA-seq datasets from Santana et al. (2023), Science, deposited under NCBI BioProject PRJNA904261. The study focuses on uncovering the transcriptional networks governing surface colonization, biofilm formation, and clinical pathogenicity in the emerging global fungal threat, *Candidozyma auris*.

By comparing the transcriptomes of the aggregative wild-type strain (AR0382_WT) with the nonaggregative wild-type strain (AR0387_WT), and comparing AR0382_WT against the insertional tnSWI1 mutant (which possesses severe surface adhesion defects), we have mapped out a critical regulatory network in *C. auris*.

2. Bioinformatics Methodology

1. Read Quality Trimming: Performed using fastp to remove low-quality bases (Phred < 30) and adapters.
2. Alignment: Reads aligned against *C. auris* strain B8441 V3 reference genome (GCA_002759435.3) using STAR utilizing built-in server indices.
3. Annotation & Quantification: Genes quantified using featureCounts against UCSC/NCBI GTF annotation files with library layout set to reverse-stranded (TruSeq protocol).
4. Statistical Testing: Pairwise Differential Gene Expression analysis executed using DESeq2 in Galaxy with significance thresholds set to FDR < 0.05 and $|\log_2 \text{Fold Change}| \geq 1.0$.

3. Global Differential Expression Summary

- Aggregative WT vs Nonaggregative WT:
 - Total Significant Differentially Expressed Genes: 195
 - Significantly Upregulated (LFC ≥ 1.0): 106
 - Significantly Downregulated (LFC ≤ -1.0): 89
- tnSWI1 Mutant vs Aggregative WT:
 - Total Significant Differentially Expressed Genes: 250
 - Significantly Upregulated (LFC ≥ 1.0): 97
 - Significantly Downregulated (LFC ≤ -1.0): 153

4. Top Regulated Adhesins and Key Genes

Below are the top 12 upregulated genes in the aggregative WT compared to the nonaggregative WT, and their corresponding regulatory response in the tnSWI1 mutant background:

Gene ID	Gene Name	WT LFC	WT FDR	Mutant LFC	Mutant FDR
B9J08_03708	SCF1	7.35	0.0	-6.82	0.0
B9J08_00369	B9J08_00369	5.40	0.0	-0.13	7.42e-02
B9J08_05200	B9J08_05200	3.65	0.0	-0.19	5.45e-03
B9J08_01084	B9J08_01084	2.54	2.99e-217	-0.55	1.47e-11
B9J08_03871	B9J08_03871	1.97	2.55e-168	0.03	7.56e-01
B9J08_05199	B9J08_05199	2.29	1.44e-102	-0.12	2.55e-01
B9J08_00453	B9J08_00453	3.29	4.14e-97	-5.36	4.06e-160
B9J08_02123	B9J08_02123	1.39	2.34e-83	-0.36	7.97e-08
B9J08_04902	B9J08_04902	1.29	8.30e-72	0.42	2.70e-15
B9J08_04747	B9J08_04747	2.66	1.72e-68	-6.98	6.50e-104
B9J08_00946	B9J08_00946	1.77	9.38e-68	-0.75	5.91e-20
B9J08_05550	B9J08_05550	1.53	2.23e-60	-0.77	1.67e-20

5. Biological Discussion & Key Insights

1. SCF1 (B9J08_03708) is a Master Aggregative Adhesin:

Our reanalysis confirms that SCF1 is highly expressed in the aggregative wild-type strain (AR0382_WT) compared to the nonaggregative strain (AR0387_WT) with a massive log2 fold change of +7.35. This confirms SCF1 is a specific and dominant surface adhesin driving cell-cell aggregation and surface biofilm colonization.

2. SWI1 is a Master Positive Transcriptional Regulator of Adhesins:

In the tnSWI1 mutant background, the expression of SCF1 completely crashes, with a log2 fold change of -6.82. Additionally, other top aggregative cell surface genes like B9J08_00453 (LFC drops by -5.36) and B9J08_04747 (LFC drops by -6.98) are completely silenced. This shows that functional SWI1 is absolutely required to activate the aggregative cell-surface program of *C. auris*. Disruption of SWI1 completely downregulates the cell-surface adhesins, which accounts for the loss of surface colonization and adhesion phenotypes in the mutant.

6. High-F

